

## Lecture 12 & 13

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Sliding mode control and its application

# Outline

- 1 Need of Terminal Sliding Modes
- 2 Terminal Sliding Mode Concept
- 3 Fast Terminal Sliding Mode
- 4 Non-Singular Terminal Sliding Mode
- 5 Prescribed Convergence Law

## Terminal Sliding Mode: Motivation

### Classical Sliding Mode Control

- Most commonly used switching manifolds are the linear hyperplanes, which guarantee asymptotic stability of the system motion during sliding mode
- System states reach the equilibrium in infinite time

### Motivation

- Need to improve the system performance during sliding mode
- System should reach equilibrium point as fast as possible
- A non-Lipschitz nonlinear manifold has faster convergence near the equilibrium point (i.e., origin)

### Terminal Sliding Mode

- In TSM, a nonlinear sliding surface is proposed
- The equilibrium is a terminal attractor, i.e., the states can be reached in finite time and are stable
- The term terminal refers to the equilibrium which can be reached in finite time and is stable

## Terminal Sliding Mode: Concept

Consider a second-order system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= f(x) + g(x)u, \quad x_1, x_2 \in \mathbb{R}\end{aligned}$$

Define the sliding variable as

$$s = x_2 + \beta x_1^{q/p}, \quad \beta > 0$$

where  $p$  and  $q$  are odd integers such that  $q < p$ . Differentiating  $s$ , we obtain

$$\begin{aligned}\dot{s} &= \dot{x}_2 + \beta \frac{q}{p} x_1^{\frac{q}{p}-1} \dot{x}_1 \\ &= f(x) + g(x)u + \beta \frac{q}{p} x_1^{\frac{q-p}{p}} x_2.\end{aligned}$$

Design the control as  $u = -g^{-1}(x) \left( f(x) + \beta \frac{q}{p} x_1^{\frac{q-p}{p}} x_2 + K \text{sign}(s) \right)$  where  $K > 0$ .

Then, simplifying further

$$\dot{s} = -K \text{sign}(s).$$

## Terminal Sliding Mode: Concept

### Reduced Order Dynamics

During the sliding mode, we achieve  $s = 0$ , that implies

$$\dot{x}_1 = x_2 = -\beta x_1^{q/p}.$$

Now solving for time  $t_1$  such that  $x_1(t) = 0$  for all  $t \geq t_1$  given any initial condition  $x_1(t_0)$  at time  $t = t_0$

$$\begin{aligned} t_1 &= t_0 - \frac{1}{\beta} \int_{x_1(t_0)}^0 x_1^{-\frac{q}{p}} dx_1 \\ &= t_0 + \frac{p}{\beta(p-q)} x_1^{\frac{p-q}{p}}(t_0). \end{aligned}$$

Note that  $p - q$  is an even number. It implies that  $x_1$  goes to zero in time  $t_1$  and remains there for all time  $t \geq t_1$ . Since  $\dot{x}_1 = 0$ , then  $x_2 = 0$  and thus, finite time stability is ensured.

## Terminal Sliding Mode: Concept

### Remarks

- The control expression contains negative exponent of  $x_1$ , so it becomes unbounded for  $x_1 = 0$
- However, during sliding, the control can be bounded by selecting  $\frac{q}{p} \in [0.5, 1)$ . It can be shown as  $u = -g^{-1}(x) \left( f(x) - \beta^2 \frac{q}{p} x_1^{2\left(\frac{q}{p} - \frac{1}{2}\right)} + K \text{sign}(s) \right)$

### Terminal Sliding Manifold

Consider the terminal sliding manifold

$$s = x_2 + \beta x_1^{q/p}$$

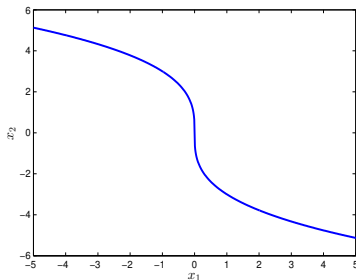


Figure: Terminal sliding manifold

## Terminal Sliding Mode: Concept

### Remarks

- TSM manifold is non-Lipschitz in nature
- Near origin, the convergence rate is much faster than the linear surface
- The solution of such a system reaches the equilibrium point in finite time
- If  $p = q$ , then TSM manifold becomes linear sliding surface

### TSM for chain of integrator system with single-input

$$\begin{aligned}\dot{x}_i &= x_{i+1} \quad i = 1, 2, \dots, n-1, \\ \dot{x}_n &= f(x) + g(x)u.\end{aligned}$$

## TSM for chain of integrator system with single-input

For the  $n^{\text{th}}$  order single-input system, hierarchical TSM manifolds are defined as

$$s_1 = \dot{s}_0 + \beta_1 s_0^{q_1/p_1}$$

$$s_2 = \dot{s}_1 + \beta_2 s_1^{q_2/p_2}$$

$$\vdots$$

$$s_{n-2} = \dot{s}_{n-3} + \beta_{n-2} s_{n-3}^{q_{n-2}/p_{n-2}}$$

$$s_{n-1} = \dot{s}_{n-2} + \beta_{n-1} s_{n-2}^{q_{n-1}/p_{n-1}}$$

where  $s_0 = x_1$ ,  $\beta_i > 0$ ,  $p_i > q_i$  and  $p_i, q_i$  are positive odd integers.

The values of the integer must satisfy following for bounded control during sliding<sup>1</sup>

$$\frac{q_k}{p_k} > \frac{n-k}{n-k+1} \quad k = n-1, \dots, 1.$$

<sup>1</sup>X. Yu and Z. Man, "Model reference adaptive control systems with terminal sliding mode", Int. J. Control, vol. 64, no. 6, pp. 1165–1176, 1996.



## TSM for chain of integrator system with single-input

### TSM Control

Differentiating  $s_{n-1}$ , we obtain

$$\begin{aligned}\dot{s}_{n-1} &= \ddot{s}_{n-2} + \beta_{n-1} \frac{q_{n-1}}{p_{n-1}} s_{n-2}^{\frac{q_{n-1}}{p_{n-1}} - 1} \frac{d}{dt} s_{n-2} \\ &= f(x) + g(x)u + \sum_{i=1}^{n-1} \frac{d^i}{dt^i} \beta_{n-i} s_{n-i-1}^{\frac{q_{n-i}}{p_{n-i}}}.\end{aligned}$$

Now, design the control law

$$u = -g^{-1}(x) \left( f(x) + \sum_{i=1}^{n-1} \frac{d^i}{dt^i} \beta_{n-i} s_{n-i-1}^{\frac{q_{n-i}}{p_{n-i}}} + K \text{sign}(s_{n-1}) \right).$$

Substituting  $u$  in the above dynamics

$$\dot{s}_{n-1} = -K \text{sign}(s_{n-1}).$$

It leads to  $s_{n-1} = 0$  in finite time. This implies  $s_{n-2} = 0$  and subsequently  $s_0 = x_1 = 0$ . Thus, all the states of the system goes to zero in finite time.

# Fast Terminal Sliding Mode Control

## Fast Terminal Sliding Mode

### Motivation

Consider the terminal sliding manifold

$$s = x_2 + \beta x_1^{q/p}.$$

During sliding  $\dot{x}_1 = -\beta x_1^{\frac{q}{p}}$ . It can be observed that

- if the initial condition is far away from the origin, the term  $x_1^{\frac{q}{p}}$  has lesser magnitude than that of linear counterpart
- convergence can be enhanced by incorporating a linear term in the terminal sliding manifold

**Fast TSM:** To achieve faster convergence, a new TSM manifold is defined as

$$s = x_2 + \alpha x_1 + \beta x_1^{q/p}$$

and during sliding  $\dot{x}_1 = -\alpha x_1 - \beta x_1^{\frac{q}{p}}$ . Thus,

- when  $x_1$  is far away from the origin  $\alpha x_1$  dominates, in other words,  $\dot{x}_1 \approx -\alpha x_1$ , so convergence is faster.
- for the values of  $x_1$  close to origin,  $\beta x_1^{\frac{q}{p}}$  will be the dominating term, so  $\dot{x}_1 \approx -\beta x_1^{\frac{q}{p}}$ .

# Fast Terminal Sliding Mode

## Reduced Order System

The reduced system during sliding can be given as

$$\dot{x}_1 = -\alpha x_1 - \beta x_1^{\frac{q}{p}}.$$

The time of convergence of  $x_1$  to zero can be obtained as

$$\begin{aligned} t_1 &= t_0 + \int_{x_1(t_0)}^0 \frac{dx_1}{-\alpha x_1 - \beta x_1^{\frac{q}{p}}} \\ &= t_0 + \int_{x_1(t_0)}^0 \frac{dx_1}{-x_1^{\frac{q}{p}} \left( \alpha x_1^{1-\frac{q}{p}} + \beta \right)} \\ &= t_0 + \frac{p}{\alpha(p-q)} \left( \ln \left( \alpha x_1^{\frac{p-q}{p}}(t_0) + \beta \right) - \ln(\beta) \right) \end{aligned}$$

where  $t_0$  is the time taken by the system to reach the fast terminal sliding manifold.

## Fast Terminal Sliding Mode

### Single-input System

$$\dot{x}_i = x_{i+1} \quad i = 1, 2, \dots, n-1,$$

$$\dot{x}_n = f(x) + g(x)u.$$

For this  $n^{\text{th}}$  order single-input system, hierarchical TSM manifolds are defined as

$$s_1 = \dot{s}_0 + \alpha_1 s_0 + \beta_1 s_0^{q_1/p_1}$$

$$s_2 = \dot{s}_1 + \alpha_2 s_1 + \beta_2 s_1^{q_2/p_2}$$

$$\vdots$$

$$s_{n-2} = \dot{s}_{n-3} + \alpha_{n-2} s_{n-3} + \beta_{n-2} s_{n-3}^{q_{n-2}/p_{n-2}}$$

$$s_{n-1} = \dot{s}_{n-2} + \alpha_{n-1} s_{n-2} + \beta_{n-1} s_{n-2}^{q_{n-1}/p_{n-1}}$$

where  $s_0 = x_1$ ,  $\beta_i > 0$ ,  $p_i > q_i$  and  $p_i, q_i$  are positive odd integers. The values of integer must satisfy for bounded control during sliding given as<sup>a</sup>

$$\frac{q_k}{p_k} > \frac{n-k-1}{n-k} \quad k = n-1, \dots, 1.$$

<sup>a</sup>X. Yu and Z. Man, "Fast terminal sliding-mode control design for nonlinear dynamical systems", IEEE Trans. Circuit Systems-1, Fundam. Theory and Appl., vol. 49, no. 2, pp. 261–264, 2002.

# Non-Singular Terminal Sliding Mode Control

## Non-Singular Terminal Sliding Mode

### Terminal Sliding Mode Control

Recall the TSM control law

$$u = -g^{-1}(x) \left( f(x) + \beta \frac{q}{p} x_1^{\frac{q}{p}-1} x_2 + K \operatorname{sign}(s) \right).$$

We see that the exponent of  $x_1$  is  $\frac{q}{p} - 1 < 0$ . So, when system trajectories cross  $x_1 = 0$  axis, then the control law becomes infinite. Such a controller can not be applied to the system and it is called singularity in the TSM.

### Non-Singular Terminal Sliding Mode

To avoid such a situation, a new terminal manifold is proposed called non-singular terminal sliding mode (NTSM)<sup>a</sup>

$$s = x_1 + \frac{1}{\beta^{\frac{p}{q}}} x_2^{\frac{p}{q}}, \quad 1 < \frac{p}{q} < 2$$

The TSM and NTSM surfaces are equivalent to each other when  $s = 0$ .

<sup>a</sup>Y. Feng, X. Yu and Z. Man, "Non-singular terminal sliding mode control of rigid manipulators", Automatica, vol. 38, no. 12, pp. 2159–2167, 2002.

## Non-Singular Terminal Sliding Mode

### Equivalence between TSM and NTSM

- It is to be noted that  $x_1^{\frac{q}{p}}$  is an odd function, i.e.,  $(-x_1)^{\frac{q}{p}} = -x_1^{\frac{q}{p}}$ .
- One way to realize this, we can take  $x_1^{\frac{q}{p}} = |x_1|^{\frac{q}{p}} \text{sign}(x_1)$ .

Now, we shall see the equivalence between TSM and NTSM when  $s = 0$ . From NTSM with  $s = 0$ , we have

$$x_1 = -\frac{1}{\beta^{\frac{p}{q}}} |x_2|^{\frac{p}{q}} \text{sign}(x_2).$$

From this, we conclude that  $\text{sign}(x_1) = -\text{sign}(x_2)$ . Multiplying both sides by  $\text{sign}(x_1)$  and then taking  $\frac{q}{p}$  power on both sides (use the fact  $|x_1| = x_1 \text{sign}(x_1)$ )

$$\beta |x_1|^{\frac{q}{p}} = |x_2|.$$

Multiplying both sides by  $\text{sign}(x_2)$ , it yields

$$-\beta |x_1|^{\frac{q}{p}} \text{sign}(x_1) = x_2.$$

This, in other words, equal to  $x_2 = -\beta x_1^{\frac{q}{p}}$ . Thus, the time taken by the system to reach  $x_1 = 0$  is the same as that of TSM.



# Non Singular Terminal Sliding Mode

Finite time reachability to NTSM manifold

Differentiating  $s$

$$\begin{aligned}\dot{s} &= \dot{x}_1 + \frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} \dot{x}_2 \\ &= x_2 + \frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} (f(x) + g(x)u).\end{aligned}$$

Design the control law as given below

$$u = -g^{-1}(x) \left( f(x) + \beta^{\frac{p}{q}} \frac{q}{p} x_2^{2-\frac{p}{q}} + K \text{sign}(s) \right).$$

Substituting for  $u$  in  $\dot{s}$ , we obtain

$$\dot{s} = -\frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} K \text{sign}(s).$$

# Non Singular Terminal Sliding Mode

## Finite time reachability to NTSM manifold

- To show convergence to origin, we consider  $V = \frac{1}{2}s^2$ . Differentiating  $V$  along the system trajectories

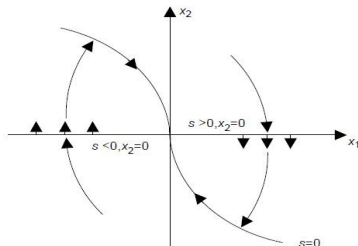
$$\dot{V} = s\dot{s} = -s \left( \frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} K \text{sign}(s) \right)$$

which, on further simplification

$$\dot{V} = -\frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} K |s|.$$

- Define  $\rho(x_2) := \frac{1}{\beta^{\frac{p}{q}}} \frac{p}{q} x_2^{\frac{p}{q}-1} K$ .
- Then  $s\dot{s} = -\rho(x_2)|s|$ . If  $x_2 \neq 0$ , we have  $\rho(x_2) > 0$ .
- That means the trajectories are attracted towards the NTSM manifold, and hence, finite time convergence is achieved.
- For  $x_2 = 0$ , we write  $\dot{x}_2 = -K \text{sign}(s)$ .

## Non Singular Terminal Sliding Mode



### Finite time reachability to NTSM manifold

- If  $s > 0$ , then  $\dot{x}_2 = -K$ . Similarly for  $s < 0$ , we have  $\dot{x}_2 = K$ .
- It implies that there exists a small vicinity  $|x_2| < \delta$  around  $x_2 = 0$  such that for  $s > 0$ , we have  $\dot{x}_2 = -K$ . Similarly for  $s < 0$ .
- Then  $x_2$  decreases for  $s > 0$  and increases for  $s < 0$ . So, the sliding trajectories will cross the boundaries  $x_2 = \delta$  and  $x_2 = -\delta$  in finite time and similarly for  $s < 0$ .
- Therefore, the trajectories are attracted towards the NTSM manifold in finite time. Thus, the proof is completed.

# Prescribed Convergence Law

## Prescribed Convergence Law

Consider a second order system

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = f(x) + g(x)u$$

where  $g(x) \neq 0$ . Define the sliding variable as  $s = x_2 + \beta|x_1|^{\frac{1}{2}}\text{sign}(x_1)$ . The control law is given as

$$u = -g^{-1}(x)(f(x) + \alpha\text{sign}(s))$$

where  $\alpha > \frac{\beta^2}{2}$ . Substituting the control in the system dynamics, we obtain

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\alpha\text{sign}(s).$$

It can be seen that trajectories are driven by a constant rate gain, hence the name prescribed convergence law<sup>ab</sup>.

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<sup>a</sup>A. Levant, "Universal single-input-single-output(SISO) sliding mode controllers with finite time convergence", IEEE Trans. Autom. Control, vol. 46, no. 9, pp. 1447–1451, 2001.

<sup>b</sup>A. Levant, "Higher-order sliding modes, differentiation and output-feedback control", Int. J. Control, vol. 76, no. 9/10, pp. 924–941, 2003.

## Prescribed Convergence Law: Proof

Differentiating  $s$  and substituting for  $u$

$$\dot{s} = -\alpha \text{sign}(s) + \frac{1}{2}\beta|x_1|^{-\frac{1}{2}}x_2.$$

It can be noted that the initial conditions may be located either in  $s > 0$  or  $s < 0$  ( $s = 0$  is trivial). Consider  $s > 0$  and then

$$\dot{x}_1 = x_2$$

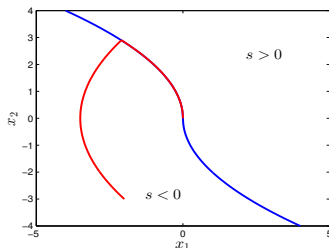
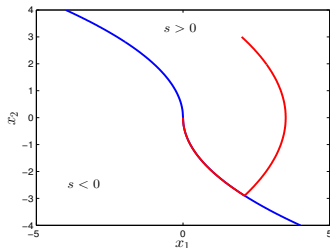
$$\dot{x}_2 = -\alpha.$$

Due to geometric reasons, the system trajectories decrease and eventually hit the curve  $s = 0$  on the way. Similarly, for the case  $s < 0$  as the dynamics takes the form

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \alpha.$$

It implies that the trajectories increase and hit  $s = 0$  in finite time.



## Prescribed Convergence Law: Proof

When  $s = 0$ , we have  $x_2 = -\beta|x_1|^{\frac{1}{2}}\text{sign}(x_1)$ , so

$$\begin{aligned}\dot{s} &= -\alpha\text{sign}(s) - \frac{1}{2}\beta^2\text{sign}(x_1) \\ &\leq -\eta\text{sign}(s).\end{aligned}$$

Thus, once the trajectories hit  $s = 0$ , it can never leave it provided  $\alpha > \frac{\beta^2}{2}$ , and hence, the sliding mode is enforced in finite time.

### System Dynamics

During sliding, we obtain  $\dot{x}_1 = -\beta|x_1|^{\frac{1}{2}}\text{sign}(x_1)$ . Consider  $V = \frac{1}{2}x_1^2$ . Then,

$$\begin{aligned}\dot{V} &= x_1\dot{x}_1 = -x_1\beta|x_1|^{\frac{1}{2}}\text{sign}(x_1) = -\beta|x_1|^{\frac{3}{2}} \\ &= -\beta 2^{\frac{3}{4}} V^{\frac{3}{4}}.\end{aligned}$$

We see that  $V$  goes to zero in time  $t_1 = t_0 + \frac{4}{\beta 2^{3/4}} V^{\frac{1}{4}}(t_0)$ . Thus, finite time stability is ensured.

# Summary

## Remarks

- Prescribed convergence law and TSM are similar except in their control structures.
- NTSM is proposed to avoid the singularity issue in TSM.
- There is no singularity in the prescribed convergence law.
- These control structures belong to second-order sliding mode control.
- We will study second-order sliding mode control in the future.



Thank You!